

ProteinChip[®] Data Manager

Software 3.0.7

Installation and Setup



Bio-Rad Laboratories, Inc.
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Table of Contents

Chapter 1: Getting Started	1	<i>Obtain a License</i>	13
<i>Welcome</i>	3	<i>Install the Database and Server Software</i>	13
<i>Software Overview</i>	3		
<i>ProteinChip Data Manager Software Components</i>	3	Chapter 3: New ProteinChip Data Manager Installation Using Oracle	17
<i>Database Software: MySQL or Oracle</i>	3	<i>Introduction</i>	19
<i>System Requirements</i>	4	<i>Before You Begin</i>	19
<i>Client-Server Configurations</i>	5	<i>Obtain a License</i>	19
<i>ProteinChip SELDI System</i>	5	<i>Oracle Database Installation</i>	19
<i>ProteinChip Biology (PBS) System II (or IIc)</i>	6	<i>Install the Database and Server Software</i>	19
<i>Installation Overview</i>	7	<i>Create the Database Using Oracle 9i</i>	19
<i>Installation Steps</i>	7	<i>Initialize the Database</i>	20
<i>What You Will Need for Installation</i>	8	<i>Install the Server Software</i>	20
<i>Installation Choices</i>	9	<i>Create the Database Tables</i>	23
<i>MySQL Users</i>	9	Chapter 4: Installation Addendum for ProteinChip Data Manager Software 3.0.7	27
<i>Oracle Users</i>	9	<i>Upgrading the Server Software From 3.0 to 3.0.7 for MySQL Users</i>	29
Chapter 2: New ProteinChip Data Manager Installation Using MySQL	11	<i>Back Up the MySQL Database</i>	29
<i>Introduction</i>	13		
<i>Before You Begin</i>	13		

<i>Uninstall the Previous Version of CiphergenExpress Software</i>	29	Chapter 7: Installing the Client Software	53
<i>Install the Database and Server Software</i>	30	<i>Before You Begin</i>	55
<i>Upgrading the Server Software From 3.0 to 3.0.7 for Oracle Users</i>	32	<i>Installing and Configuring the Client Software</i>	55
<i>Back Up the Oracle Database</i>	32	<i>Installing the Client Software</i>	55
<i>Uninstall the Previous Version of CiphergenExpress Software</i>	32	<i>Configure the Client Software's Server Connection</i>	56
<i>Install the Server Software</i>	32	<i>Register the Server</i>	58
Chapter 5: Upgrading From Version 2.x Using MySQL	37	Chapter 8: Troubleshooting	59
<i>Introduction</i>	39	<i>Troubleshooting Installations and Upgrades</i>	61
<i>Before you begin</i>	39	<i>Contacts</i>	61
<i>Back Up the MySQL Database</i>	39	Index	63
<i>Uninstall the Previous Version of CiphergenExpress Software</i>	40		
<i>Upgrade the Server and Database</i>	40		
<i>Install the Server Software</i>	40		
<i>Install the Database</i>	43		
Chapter 6: Upgrading From Version 2.x Using Oracle	45		
<i>Introduction</i>	47		
<i>Before You Begin</i>	47		
<i>Back Up the Oracle Database</i>	47		
<i>Uninstall the Previous Version of CiphergenExpress Software</i>	48		
<i>Upgrade the Server and Database</i>	48		
<i>Install the Server Software</i>	48		
<i>Update the Database Tables</i>	51		

Chapter 1: Getting Started

Welcome	3
Software Overview	3
ProteinChip Data Manager Software Components	3
Database Software: MySQL or Oracle	3
System Requirements	3
Client-Server Configurations	5
ProteinChip SELDI System	5
ProteinChip Biology (PBS) System II (or IIc)	6
Installation Overview	7
Installation Steps	7
What You Will Need for Installation	8
Installation Choices	8

Welcome

Welcome to the installation guide for ProteinChip data manager software. This booklet provides complete installation and upgrade instructions for ProteinChip data manager software, together with additional setup and administrative tasks.

Software Overview

ProteinChip Data Manager Software Components

ProteinChip data manager software is a client-server application, and is comprised of three components: client software, server software, and a MySQL or Oracle relational database. The database is where the trace data, sample properties, peak lists, *etc.* are stored. The server software performs most of the data processing. Commands issued through the client software are received by the server software for execution. The client software provides the user graphical access to the server software, and to ProteinChip SELDI system instruments.

Database Software: MySQL or Oracle

The server software is designed to use MySQL or Oracle 9i as the database software.

- MySQL database software is included on the ProteinChip data manager software CD, and a MySQL license is provided for the initial installation
- Oracle users are responsible for obtaining Oracle 9i software and a license

Refer to Chapter 2 for instructions on installing the server in conjunction with a MySQL server. If you wish to use an Oracle database instead of the MySQL database, please refer to Chapter 3, New ProteinChip Data Manager Installation Using Oracle, for installation information.

System Requirements

The database and server software should be installed on a Windows computer that can function as a server. The resource recommendations for the server and client computers are outlined in Table 1.1 and Table 1.2; however, processing and data storage requirements depend greatly on the number of samples and spectra stored and processed as well as the number of concurrent users to support.

ProteinChip data manager server software must be installed on a computer that is a dedicated server for ProteinChip Data manager software, with no other server or database applications running on that machine. This requirement is necessitated by the large amount of system resources involved in processing and analyzing SELDI data.

The recommended computer settings are specified for users who wish to cluster and visualize more than one million peaks in one analysis.

Table 1.1: System requirements for the server software and the database.

Component	Required	Recommended
Operating system	Microsoft Windows 2000 Server or Microsoft Windows 2000 or Windows XP	Microsoft Windows 2000 Server or Microsoft Windows 2000 or Windows XP
Processor	1 GHz	2 GHz or greater
Memory	1 GB RAM	2 GB RAM or greater
Free disk space	50 GB	100 GB or greater
Database	MySQL 4.0 (included) or Oracle 9i release 1 or later (to be obtained by customer)	MySQL 4.0 (included) or Oracle 9i release 1 or later (to be obtained by customer)
Misc.	CD-ROM for program installation, network connectivity	CD-ROM for program installation, network connectivity

Table 1.2: Client workstation system requirements for ProteinChip data manager client software.

Component	Required	Recommended
Operating system	Microsoft Windows 2000 or Windows XP	Microsoft Windows 2000 or Windows XP
Processor	500 MHz	1 GHz or greater
Memory	256 MB RAM	1 GB RAM or greater
Free disk space	100 MB	1 GB or greater
Misc.	Network connection to the ProteinChip data manager server computer; minimum 1024 x 768 screen resolution	Network connection to ProteinChip data manager server computer; minimum 1024 x 768 screen resolution

**NOTE**

*During the first use of ProteinChip data manager software, the server software must be registered. A license file is required to register the server software, and can be obtained by sending an email containing the server computer name and IP address to LSG_TECHSERV_US@bio-rad.com. The license is installed by selecting **Help | Register** from the client software menu.*

Client-Server Configurations

There are two different client-server configurations possible, depending on the type of instrument you have. Please identify the appropriate configuration for your instrument and refer to the appropriate section of the instrument manual for setup information.

ProteinChip SELDI System

The ProteinChip SELDI system is configured with a computer that contains both the ProteinChip data manager server and database, and also serves as the primary workstation. Additional client workstation computers can be networked to the instrument (Fig. 1.1).

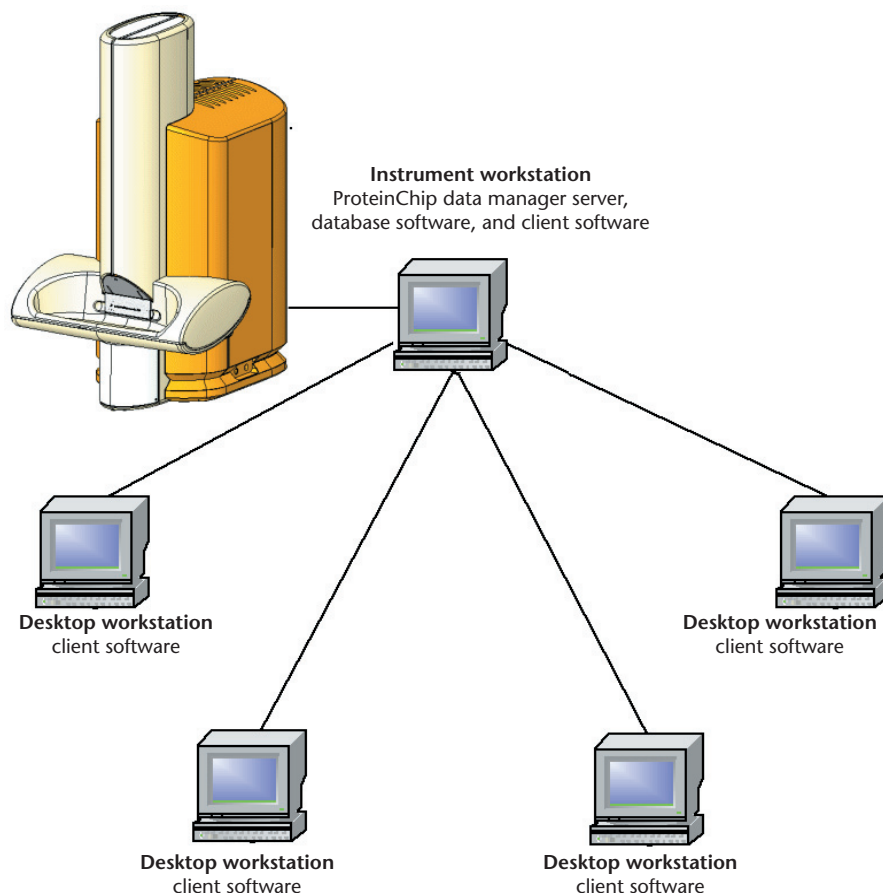


Fig. 1.1: Client-server configuration for the ProteinChip SELDI System.

ProteinChip Biology (PBS) System II (or IIc)

If you purchased ProteinChip Data manager software for use with a PBS-II or PBS-IIc ProteinChip reader, you will need to install and set up the software. Before installing the client software, the server and database software must be installed on a networked dedicated server computer by a network administrator. You must also install ProteinChip software 3.2.1 or later to communicate properly with ProteinChip data manager software (Fig. 1.2).

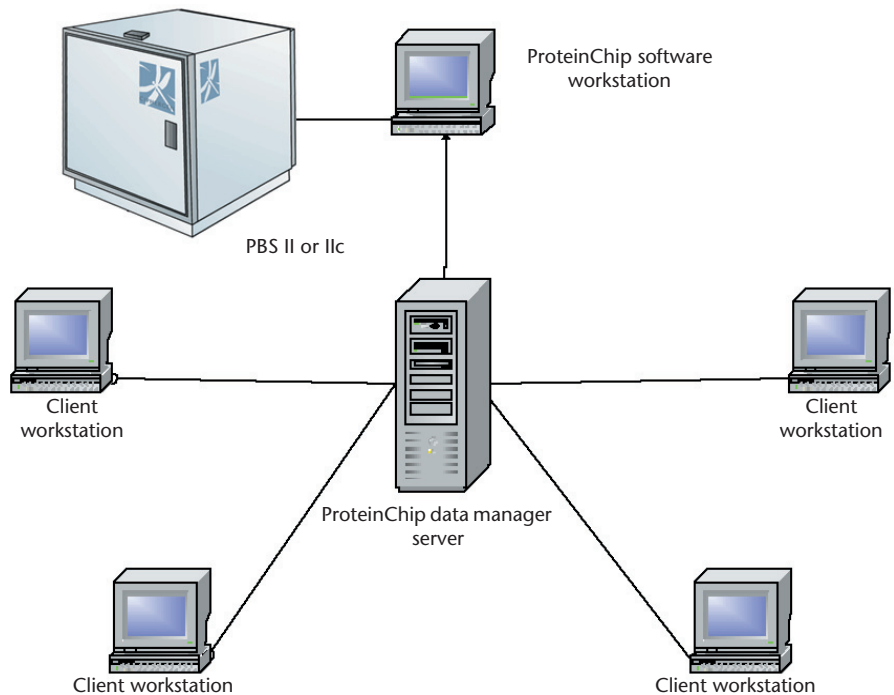


Fig. 1.2: Client-server configuration for the ProteinChip biology system.

If you are using ProteinChip Data manager software with a PBS-II or Ilc, you will need to configure ProteinChip software to communicate with the ProteinChip data manager server after the server software and database are installed on the server computer. See the ProteinChip Software Operator Manual for instructions on how to configure it.

Installation Overview

Installation Steps

Installation of ProteinChip data manager software comprises six steps:

1. **Set up the server computer on the network.** This step should be performed by an IT person. It involves connecting the computer to the network and assigning a server hostname and IP address. We recommend obtaining a static IP address for the ProteinChip data manager server computer.

2. **Obtain a license file.** In order to use ProteinChip data manager software, a valid license file must be obtained from Bio-Rad Laboratories, Inc. The license file required for registration can be obtained by sending an email containing your name, company, server hostname, and server IP address to: *LSG_TECHSERV_US@bio-rad.com*. If you are purchasing a server in conjunction with ProteinChip Data manager software, your server will be preinstalled with ProteinChip Data manager software and will be prelicensed.
3. **Install the database and server software.** This step should be performed by an IT person on the computer to be used as the server. The installation must be performed using an account that has administrative privileges on the computer.
4. **Install the client software.** The client setup program is downloaded from the ProteinChip data manager server using an Internet browser.
5. **Register the ProteinChip data manager server software.** Using the license file obtained in the second step, the server is registered and readied for use. Note that if you have purchased a server from Bio-Rad, it will be preinstalled and prelicensed in the factory.
6. **Set up instrument communication.** If you are using a ProteinChip SELDI System you must configure the instrument directly via the client software, which also serves as the controlling software.

What You Will Need for Installation

1. A ProteinChip data manager software CD.
2. A networked computer to be used as the ProteinChip data manager server with at least 50 GB of free disk space. The software must be installed using an account with administrative access to the computer.
3. At least one workstation computer to run the client software, with at least 100 MB of free disk space. This workstation must reside on the network and have an Internet browser such as Microsoft Internet Explorer installed on it.
4. Your ProteinChip data manager license file, obtained by sending an email to *LSG_TECHSERV_US@bio-rad.com*.

Installation Choices

Installation procedures are broken down into chapters by system configuration. The number 2.x refers to both version 2.0 and 2.1 of ProteinChip data manager software. You cannot upgrade directly from ProteinChip data manager software version 1.0 to version 3.0.7.

Use the list below to choose the appropriate chapters to use for installing ProteinChip data manager on your network and workstation(s). No matter which chapter is used as a guide for installing the database and server software, you will need to follow Chapter 7, Installing the Client Software, to install ProteinChip data manager client software on the workstation(s).

MySQL Users

- **New version 3.0.7 installation** — Chapter 2, New ProteinChip Data Manager Installation Using MySQL, and Chapter 7, Installing the Client Software.
- **Upgrading from version 2.x** — Chapter 5, Upgrading From Version 2.x Using MySQL, and Chapter 7, Installing the Client Software.

Oracle Users

- **New version 3.0.7 installation** — Chapter 3, New ProteinChip Data Manager Installation Using Oracle, and Chapter 7, Installing the Client Software.
- **Upgrading from version 2.x** — Chapter 6, Upgrading From Version 2.x Using Oracle, and Chapter 7, Installing the Client Software.

Chapter 2: New ProteinChip Data Manager Installation Using MySQL

Introduction	13
Before You Begin	13
Obtain a License	13
Install the Database and Server Software	13

Introduction

This section provides instructions for creating a **new** installation of ProteinChip data manager software using a MySQL database. It does not include instructions for updating from version 2.x. If you are updating from an earlier version, or using an Oracle database, please see Installation Choices on page 9. Please email any installation questions to LSG_TECHSERV_US@bio-rad.com.

Before You Begin

Obtain a License

ProteinChip data manager software must be registered to function. A license file is required to register the software, and can be obtained by sending an email containing the server computer name, IP address, and company name to LSG_TECHSERV_US@bio-rad.com.

Install the Database and Server Software

The MySQL database software is included on the ProteinChip data manager software CD and will be installed and configured during the installation.

1. Insert the ProteinChip data manager software CD into the CD-ROM drive of the computer to be used as the ProteinChip data manager server. If the program does not start automatically after a few moments, browse to the CD using Windows Explorer and double-click *setup.exe*.
2. At the **Welcome** page, click the **Next** button.
3. At the **License Agreement** page, please read the license agreement. You must click the **Yes** button to accept the agreement in order to proceed.
4. Please read the preinstall instructions on the **Install checklist** page, then click the **Next** button.
5. At the **Choose Destination Location** page, specify where on the hard drive you would like to install the server software. The default path is C:\PCDM. If you must change the path (to change the drive letter, for example) make sure there are no spaces in the path, or the software will not function correctly. Click the **Next** button to continue.



IMPORTANT

*If you modify any part of the default path, you **must** ensure that there are **no spaces** in the path name or the software will not run correctly.*

6. At the **Setup Type** page, select **1. New installation using a MySQL database** and then click the **Next** button (Fig. 2.1).

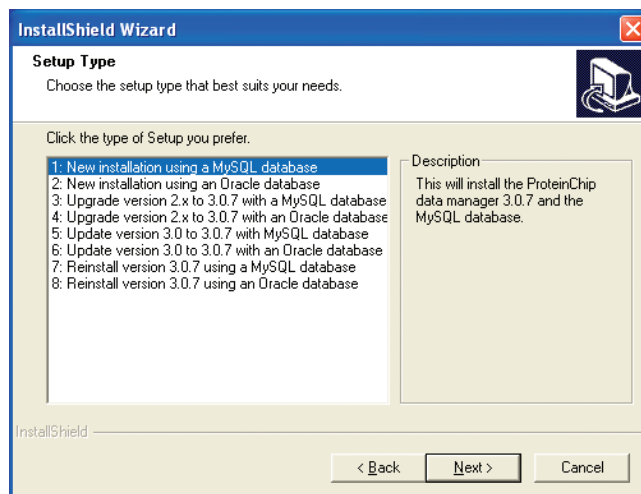


Fig. 2.1: The Setup Type page.

7. At the **Select Program Folder** page, choose the location where the program shortcuts will be placed in the computer's **Start** menu. The default location is **ProteinChip Data Manager Software version 3.0.7**. Click the **Next** button to continue.
8. The **Installation Summary** page (Fig. 2.2) displays summary information about the setup type that was chosen. Please review the contents before proceeding. For new installations using MySQL, the summary should show your specified installation directory, and **New installation with a MySQL database** for the installation type. If any of the information is incorrect, click the **Back** button to make corrections. If the information is accurate, click the **Next** button to begin the installation.

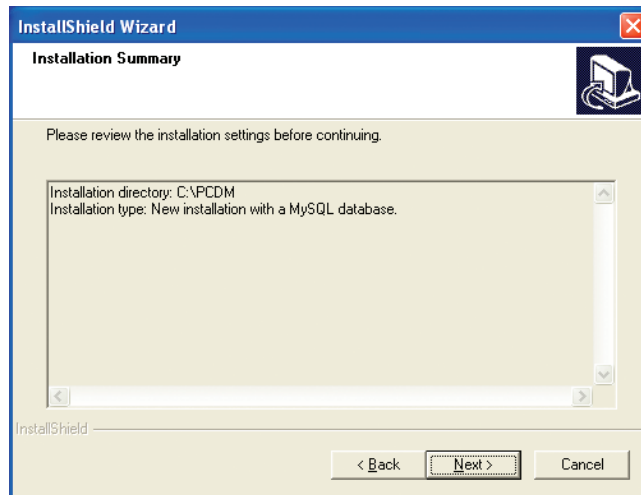


Fig. 2.2: The Installation Summary page.

9. Once the server installation is complete, you are ready to install the client software onto the workstation. Proceed to Chapter 7, Installing the Client Software for further instructions.



NOTE

If any problems or errors were encountered during the installation, please send an email, including any error messages and attached screenshots, to LSG_TECHSERV_US@bio-rad.com.

Chapter 3: New ProteinChip Data Manager Installation Using Oracle

Introduction	19
Before You Begin	19
Obtain a License	19
Oracle Database Installation	19
Install the Database and Server Software	19
Create the Database Using Oracle 9i	19
Initialize the Database	20
Install the Server Software	20
Create the Database Tables	23

Introduction

This section provides instructions for creating a **new** installation of ProteinChip data manager software using an Oracle database. It does not include instructions for updating from version 2.x. If you are updating from version 2.x, or using a MySQL database, please see Installation Choices on page 8.

Please email any installation questions to LSG_TECHSERV_US@bio-rad.com.

Before You Begin

Obtain a License

ProteinChip data manager software must be registered to function. A license file is required to register the software, and can be obtained by sending an email containing the server computer name, IP address, and company name to LSG_TECHSERV_US@bio-rad.com.

Oracle Database Installation

If you will be using an Oracle database with ProteinChip data manager software, a database administrator must install the Oracle database before configuring the database schema and installing the server software. The Oracle database installation and initialization procedures should be reviewed and performed by an Oracle database administrator or personnel with equivalent experience. These procedures should only be performed for a **new** ProteinChip data manager installation using an Oracle database.



NOTE

Your organization is responsible for obtaining the Oracle database software and an Oracle license.

Install the Database and Server Software

These procedures should only be performed for a **new** ProteinChip data manager software installation using an Oracle database. The Oracle database install and initialize procedures should be reviewed and performed by an Oracle database administrator or personnel with equivalent experience.

Create the Database Using Oracle 9i

These procedures should only be performed for a **new** ProteinChip data manager software installation using an Oracle database.

1. Launch the **Oracle Database Configuration Assistant**.
2. On the **Operations** page, select **Create a Database** and click **Next**.

3. On the **Templates** page, select **Data Warehouse** and click **Next**.
4. On the **Database Identification** page, enter *cipherdb* followed by the domain name (e.g., *cipherdb.bio-rad.com*) for the global database name. The **SID** should be *cipherdb* and should be entered automatically by the installation software.
5. Click **Finish**. The default values will be used for all subsequent settings.
6. Click **OK** in the **Summary** dialog box. The database setup should begin.
7. Once the setup completes, click **Exit** in the database configuration assistant window.

Initialize the Database

These procedures should only be performed for a **new** ProteinChip data manager software installation using an Oracle database.

1. Copy the *setupdb.sql* and *execsql.cmd* files from the ProteinChip data manager software installation CD to the Oracle root folder. These files are located in the *Oracle_setup* folder on the CD.
2. Edit the *setupdb.sql* file using a text editor such as Notepad. Replace the C:\oracle portion of the path on the first line with the actual Oracle root directory and save the file.
3. Edit the *execsql.cmd* file using a text editor such as Notepad. Replace the system password with your password (by default, "manager") and save the file.
4. Run *execsql.cmd*.

Install the Server Software

1. Insert the ProteinChip data manager software CD into the CD-ROM drive of the computer to be used as the server. If the program does not start automatically after a few moments, browse to the CD using Windows Explorer and double-click *setup.exe*.
2. At the **Welcome** page, click the **Next** button.
3. At the **License Agreement** page, please read the license agreement. You must click the **Yes** button to accept the agreement in order to proceed.
4. Please read the preinstall instructions on the **Install checklist** page, then click the **Next** button.
5. At the **Choose Destination Location** page, specify where on the hard drive you would like to install the server software. The default path is C:\PCDM. If you must change the path (to change the drive letter, for example), make sure there are no spaces in the path, or the software will not function correctly. Click the **Next** button to continue.

**IMPORTANT**

*If you modify any part of the default path, you **must** ensure that there are **no spaces** in the path name or the software will not run correctly.*

- At the **Setup Type** page, select **2. New installation using an Oracle database** and then click the **Next** button (Fig. 3.1).

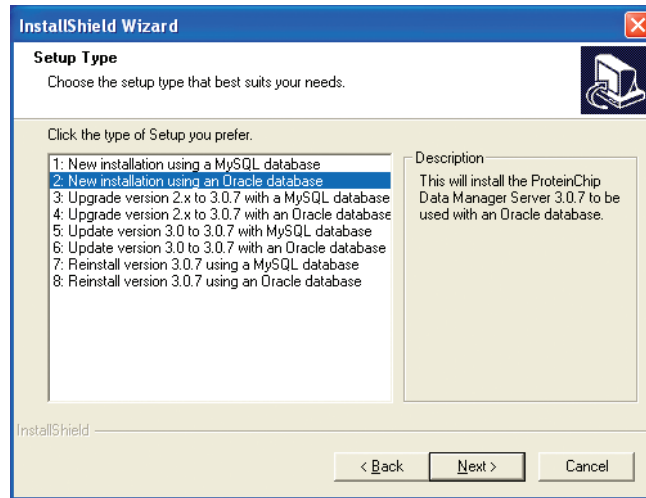


Fig. 3.1: The **Setup Type** page.

- At the **Enter Text** page (Fig. 3.2), enter in the network name of the computer that is running the Oracle database. If server software and the Oracle database are running on the same machine, this name can be left as "localhost". Click the **Next** button to continue.

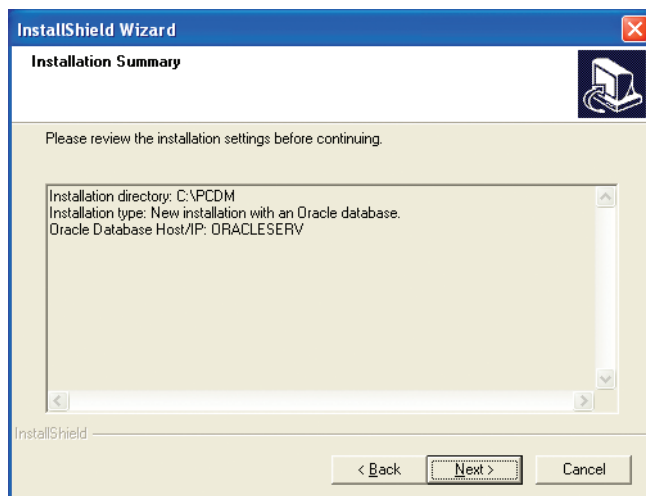


Fig. 3.2: The Enter Text page.

8. At the **Select Program Folder** page, choose the location where the program shortcuts will be placed on the computer's **Start** menu. The default location is **ProteinChip Data Manager Software version 3.0.7**. Click the **Next** button to continue.
9. The **Installation Summary** page (Fig. 3.3) displays summary information about the setup type that was chosen. Please review the contents before proceeding. For new installations using Oracle, the summary should show the specified installation directory, and **New installation using an Oracle database** for the installation type. Additionally, the **Host** for the Oracle computer should be displayed. If any of the information is incorrect, click the **Back** button to make corrections. If the information is accurate, click the **Next** button to begin the installation. Once the files are copied, another installation program will be launched to create the database tables.

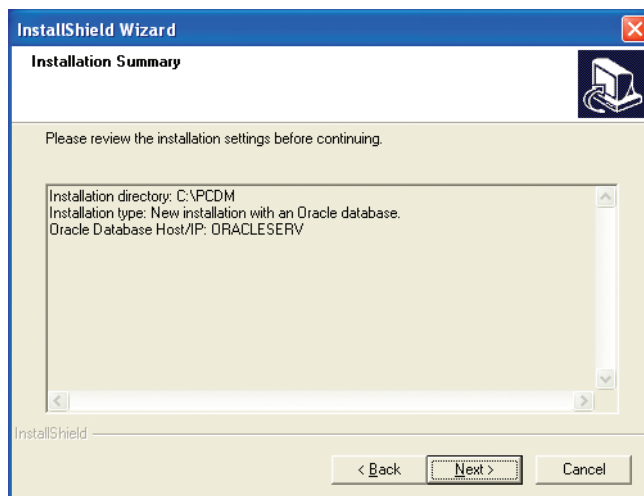


Fig. 3.3: The Installation Summary page.

Create the Database Tables

A separate, embedded installation program is used to create the Oracle database tables.

1. Once the program files have been transferred, the database installer program will launch automatically. On the first page of the **ProteinChip Data Manager Database Installer**, under **Choose install type**, select **Initialize ProteinChip Data Manager Oracle version 3.0**, then click the **Next** button (Fig. 3.4).

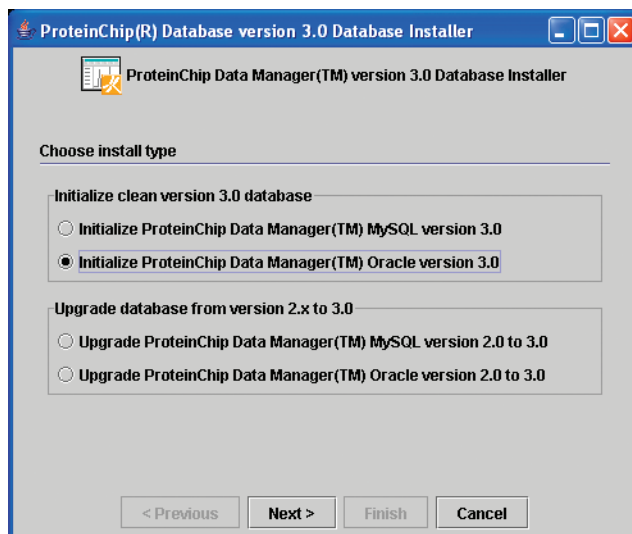


Fig. 3.4: The first page of the ProteinChip Data Manager Database Installer.

2. On the next page of the **ProteinChip Data Manager Database Installer** (Fig. 3.5), under **Enter Oracle connection information**, enter the following information:
 - a. Enter the hostname or IP address of the computer that contains the Oracle database ("localhost" can be used if the database is on the same machine upon which you are installing the server) into the **Host/IP** field.
 - b. The **SID/Database** should be *cipherdb* as outlined in Create the Database Using Oracle 9i on page 19.
 - c. The **User** should be *ciphergen*.
 - d. Enter the **Password** for the database. The default password is *express*.Once all the information is entered, click the **Next** button to continue.

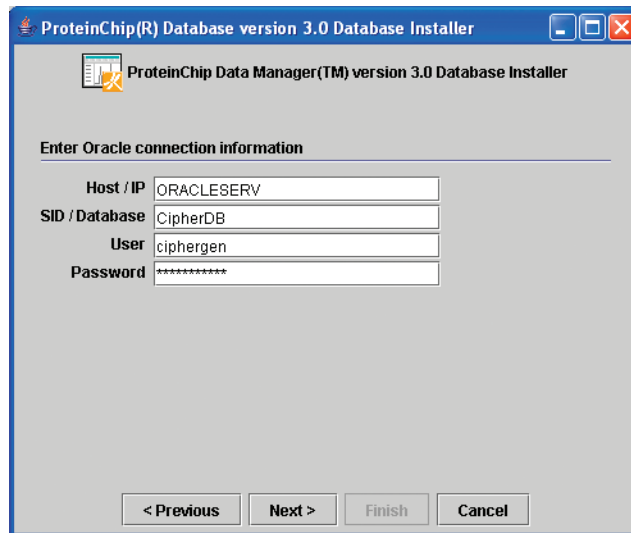


Fig. 3.5: The second page of the **ProteinChip Data Manager Database Installer**.

3. The **ProteinChip Data Manager Install Action** window displays the status of the database table creation. Once the process is complete, click **Finish**.
4. Once the database installation is complete, you will be returned to the main installer program. Click the **Finish** button to complete the installation.
5. After the server installation is complete, you are ready to install the client software onto the workstation. Proceed to Chapter 7, *Installing the Client Software*, for further instructions.



NOTE

If any problems or errors were encountered during the installation, please send an email, including any error messages and attached screenshots, to LSG_TECHSERV_US@bio-rad.com.

Chapter 4: Installation

Addendum for ProteinChip Data Manager Software 3.0.7

Upgrading the Server Software From 3.0 to 3.0.7 for MySQL Users	29
Back Up the MySQL Database	29
Uninstall the Previous Version of CiphergenExpress Software	29
Install the Database and Server Software	30
Upgrading the Server Software From 3.0 to 3.0.7 for Oracle Users	32
Back Up the Oracle Database	32
Uninstall the Previous Version of CiphergenExpress Software	32
Install the Server Software	32

Upgrading the Server Software From 3.0 to 3.0.7 for MySQL Users

This section provides instructions for upgrading ProteinChip data manager software using a MySQL database from version 3.0 to version 3.0.7. The upgrade process installs the new server software. You must also install a new version of the ProteinChip data manager client software.



NOTE

The upgrade procedure must be performed on the server computer itself, and cannot be performed from a client workstation.

Back Up the MySQL Database

Ciphergen strongly recommends that you back up the database before performing the upgrade procedure.

1. Stop the MySQL service by selecting **Control Panel | Administrative Tools | Services**. Right-click on **MySQL** and select **Stop** in the pop-up menu.
2. Stop the **CiphergenExpress Server** service by selecting **Control Panel | Administrative Tools | Services**. Right-click on **CiphergenExpress Server** and select **Stop** from the pop-up menu.
3. Once the services are stopped, locate the folder containing the ProteinChip data manager database. The default folder location is C:\CEDM\mysql. Within this folder, locate the subfolder named **data**.
4. Make a backup copy of the **data** folder in another location, preferably on another computer. This may take some time, depending on the size the folder.



NOTE

*Leave the original **data** folder in its original location. It is required for the remaining steps of the update procedure.*

Uninstall the Previous Version of CiphergenExpress Software

Before ProteinChip data manager software 3.0.7 can be installed, you must first uninstall the previous version.

1. Uninstall the CiphergenExpress server software. Select **Control Panel | Add/Remove Programs**. Locate the program called **CiphergenExpress Server version 3.0** and uninstall it.

**IMPORTANT**

*Do not delete the **CEDM** folder! This folder contains all the data to be transferred to the new version.*

Install the Database and Server Software

The MySQL database software is included on the ProteinChip data manager software CD and will be installed and configured during the installation.

1. Insert the ProteinChip data manager software 3.0.7 CD into the CD-ROM drive of the computer to be used as the ProteinChip data manager server. If the program does not start automatically after a few moments, browse to the CD using Windows Explorer and double-click *setup.exe*.
2. In the **Welcome** page, click the **Next** button.
3. At the **License Agreement** page, please read the license agreement. You must click the **Yes** button to accept the agreement in order to proceed.
4. Please read the preinstall instructions on the **Install checklist** page, then click the **Next** button.
5. In the **Choose Destination Location** page, specify the original location of the previous server installation. The default path is C:\CEDM. Click the **Next** button to continue.
6. In the **Setup Type** page, select **5: Upgrade version 3.0 to 3.0.7 with a MySQL database** and then click the **Next** button (Fig. 4.1).

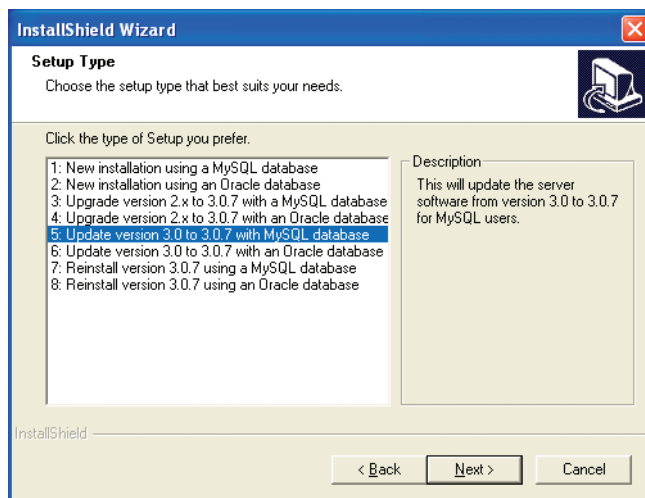


Fig. 4.1: The Setup Type page.

7. In the **Select Program Folder** page, choose the location where the program will be installed (a shortcut will also be placed in the corresponding location in the Windows **Start** menu). The default location is **ProteinChip Data Manager Server version 3.0.7**. When finished, click the **Next** button to continue.
8. The **Installation Summary** page displays information about the selected setup type (Fig. 4.2). Please review the information before proceeding. The summary should display the installation directory and **Upgrade from 3.0 to 3.0.7 with a MySQL database** for the installation type. If any of the information is incorrect, click the **Back** button to make corrections. If the information is accurate, click the **Next** button to begin the installation.

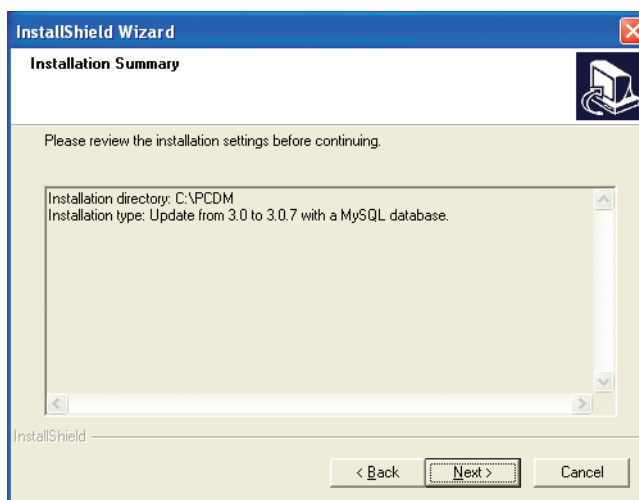


Fig. 4.2: The Setup Type page.

9. Once the server software installation is complete, you are ready to install the ProteinChip data manager client software onto the workstation. Proceed to Chapter 7, Installing the Client Software, for further instructions.



NOTE

If any problems or errors were encountered during the installation, please send an email, including any error messages and attached screenshots, to LSG_TECHSERV_US@bio-rad.com.

Upgrading the Server Software From 3.0 to 3.0.7 for Oracle Users

This section provides instructions for upgrading ProteinChip data manager software using an Oracle database from version 3.0 to version 3.0.7. The upgrade process installs the new server software. You must also install a new version of the client software.



NOTE

The upgrade procedure must be performed on the server computer itself, and cannot be performed from a client workstation.

Back Up the Oracle Database

Bio-Rad strongly recommends that you back up the database before performing the upgrade procedure. Consult Oracle technical documentation for backup procedures if your database is implemented with Oracle.



NOTE

*You may need to stop the **CiphergenExpress Server** service before performing a backup. To stop the service, select **Control Panel | Administrative Tools | Services**. Right-click **CiphergenExpress Server** and select **Stop** from the pop-up menu.*

Uninstall the Previous Version of CiphergenExpress Software

Before ProteinChip data manager software 3.0.7 can be installed, you must first uninstall the previous version.

1. Select **Control Panel | Add/Remove Programs**. Locate the program called **CiphergenExpress Data Manager Server** and uninstall it.

Install the Server Software

1. Insert the ProteinChip data manager software version 3.0.7 CD into the CD-ROM drive of the computer to be used as the ProteinChip data manager server. If the program does not start automatically after a few moments, browse to the CD using Windows Explorer and double-click *setup.exe*.
2. At the **Welcome** page, click the **Next** button.
3. At the **License Agreement** page, please read the license agreement. You must click the **Yes** button to accept the agreement in order to proceed.
4. Please read the preinstall instructions on the **Install checklist** page, then click the **Next** button.

5. In the **Choose Destination Location** page, specify the original location of the previous server installation. The default path is C:\PCDM. Click the **Next** button to continue.
6. At the **Setup Type** page, select **6: Upgrade version 3.0 to 3.0.7 with an Oracle database** and then click the **Next** button (Fig. 4.3).

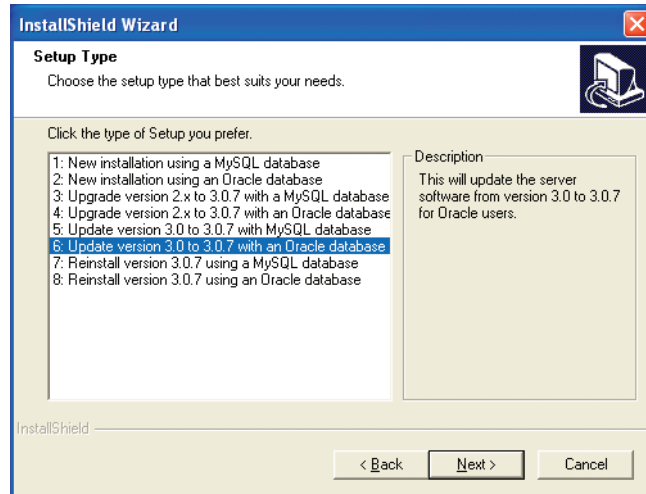


Fig. 4.3: The Setup Type page.

7. At the **Enter Text** page, enter in the network name of the computer that is running the Oracle database. If the server software and the Oracle database are running on the same machine, this field can be left as "localhost". Click the **Next** button to continue (Fig. 4.4).

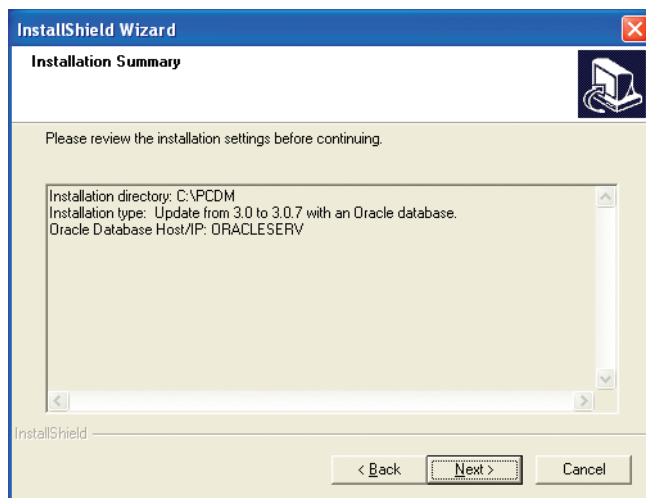


Fig. 4.4: The Enter Text page.

8. In the **Select Program Folder** page, choose the location where the program will be installed (a shortcut will also be placed in the corresponding location in the Windows **Start** menu). The default location is **ProteinChip Data Manager Software version 3.0.7**. When finished, click the **Next** button to continue.
9. The **Installation Summary** page displays information about the selected setup type. Please review the information before proceeding. The summary should display the installation directory and **Upgrade from 3.0 to 3.0.7 with an Oracle database** for the installation type. If any of the information is incorrect, click the **Back** button to make corrections. If the information is accurate, click the **Next** button to begin the installation (Fig. 4.5).

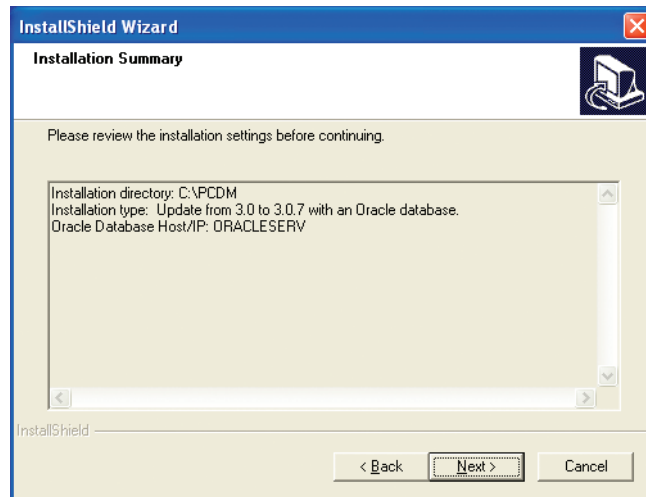


Fig. 4.5: The Installation Summary page.

10. Once the server software installation is complete, you are ready to install the ProteinChip data manager client software onto the workstation. Proceed to Chapter 7, Installing the Client Software, for further instructions.



NOTE

If any problems or errors were encountered during the installation, please send an email, including any error messages and attached screenshots, to LSG_TECHSERV_US@bio-rad.com.

Chapter 5: Upgrading From Version 2.x Using MySQL

Introduction	39
Before you begin	39
Back Up the MySQL Database	39
Uninstall the Previous Version of CiphergenExpress Software	40
Upgrade the Server and Database	40
Install the Server Software	40
Install the Database	43

Introduction

This chapter provides instructions for upgrading CiphergenExpress Server software version 2.x implemented with a MySQL database to ProteinChip Data Manager Server version 3.0.7. You cannot upgrade directly from CiphergenExpress Server version 1.0 to version 3.0.7. The upgrade process installs the new server software and updates the database tables to be compatible with new features included in the version 3.0.7 software. If you are not updating from version 2.x using a MySQL database, please see Installation Choices on page 8.

Please email any installation questions to LSG_TECHSERV_US@bio-rad.com.



NOTE

This upgrade can take several hours if the database is very large.



IMPORTANT

*The upgrade process requires free disk space equal to a quarter the size of the database. The current size of the database is found in the **Properties** dialog box for the MySQL data folder. By default, this folder is in the C:\CiphergenExpress_Database folder.*



WARNING

Follow the instructions below exactly. Failure to do so may result in a failure to migrate the existing data. Email LSG_TECHSERV_US@bio-rad.com if you have any questions.

Before you begin

Back Up the MySQL Database

Bio-Rad strongly recommends that you back up the version 2.x database before performing the upgrade procedure.

1. Stop the MySQL service by selecting **Control Panel | Administrative Tools | Services**. Right-click on **MySQL** and select **Stop** in the pop-up menu.
2. Stop the **CiphergenExpress Server** service by going to **Control Panel | Administrative Tools | Services**. Right-click on **CiphergenExpress Server** and select **Stop** from the pop-up menu.
3. Once the services are stopped, locate the folder containing the MySQL database. The default folder location is: C:\CiphergenExpress_Database. Within this folder, locate the subfolder named **data**.

4. Make a copy of the **data** folder in another location, preferably on another computer. This may take some time, depending on the size the folder.
5. After the folder has finished copying, be sure to close the **Services** dialog box — leaving it open can interfere with the installation process.

**NOTE**

*Leave the original **data** folder in its original location. It is required for the remaining steps of the update procedure.*

Uninstall the Previous Version of CiphergenExpress Software

Before ProteinChip data manager software 3.0.7 can be installed, you must first uninstall the previous version.

1. Uninstall the CiphergenExpress server software. Select **Control Panel | Add/Remove Programs**. Locate the program called **CiphergenExpress Server** and uninstall it. Delete the **CiphergenExpress_Server** folder.
2. Uninstall the CiphergenExpress database program. Select **Control Panel | Add/Remove Programs**. Locate the program called **CiphergenExpress Database** and uninstall it. (You may receive an error saying that the uninstaller could not delete the “data” directory. This is expected).

**IMPORTANT**

*Do not delete the **CiphergenExpress Database** folder! This folder contains the data to be transferred to the new version.*

3. Select **Control Panel | Add/Remove Programs**. Uninstall **Java 2 Runtime Environment, SE v1.4.2**. Then, uninstall **Java 2 SDK, SE v1.4.2**.

Upgrade the Server and Database

Install the Server Software

The MySQL database software is included on the ProteinChip data manager software CD and will be installed and configured during the installation.

1. Insert the ProteinChip data manager software CD into the CD-ROM drive of the computer to be used as the ProteinChip data manager server. If the program does not start automatically after a few moments, browse to the CD using Windows Explorer and double-click *setup.exe*.
2. At the **Welcome** page, click the **Next** button.

3. At the **License Agreement** page, please read the license agreement. You must click the **Yes** button to accept the agreement in order to proceed.
4. Please read the preinstall instructions on the **Install checklist** page, then click the **Next** button.
5. At the **Choose Destination Location** page, specify where on the hard drive you would like to install the ProteinChip data manager server software. The default path is C:\PCDM. If you must change the path (to change the drive letter, for example), make sure there are no spaces in the path or the software will not function correctly. Click the **Next** button to continue.

**IMPORTANT**

*If you modify any part of the default path, you **must** ensure that there are **no spaces** in the path name or the software will not run correctly.*

6. At the **Setup Type** page, select **3: Upgrade from v.2x to v3.0.7 with a MySQL database** and then click the **Next** button (Fig. 5.1).

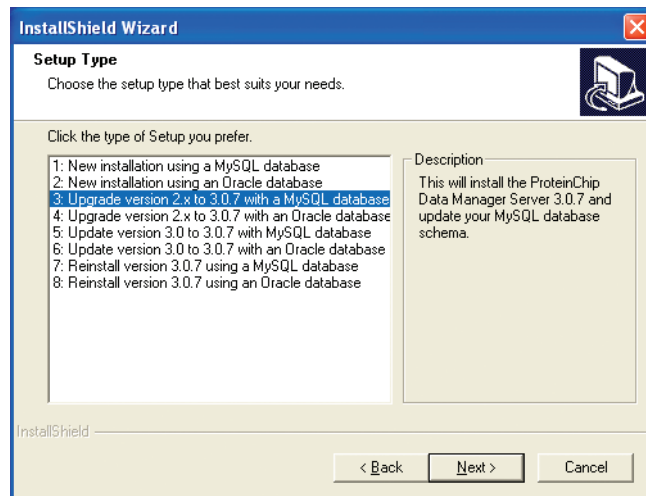


Fig. 5.1: The **Setup Type** page.

7. The installer will prompt you to confirm the upgrade (Fig. 5.2). Click the **Yes** button to continue.

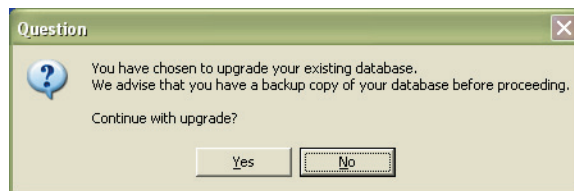


Fig. 5.2: The confirmation dialog box.

8. After confirming the upgrade path, you will be prompted to browse to the location of the original 2.x database (Fig. 5.3). The default location for the database in version 2.x was C:\CiphergenExpress_Database.

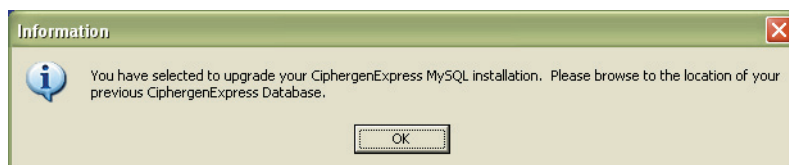


Fig. 5.3: The browse prompt dialog box.

9. If the installer successfully determines that the folder contains a valid database, the database location will be displayed as shown in Fig. 5.4. Click **OK** to proceed.

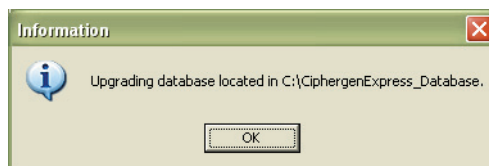


Fig. 5.4: The upgrade location dialog box.

10. At the **Select Program Folder** page, choose the location where the program shortcuts will be placed on the computer's **Start** menu. The default location is **ProteinChip Data Manager Server version 3.0.7**. Click the **Next** button to continue.
11. The **Installation Summary** page (Fig. 5.5) displays summary information about the setup type that was chosen. Please review the contents before proceeding. For upgrades using MySQL, the summary should show the specified installation directory and **Upgrade from 2.x to 3.0.7 with a**

MySQL database for the installation type. The directory of data to be upgraded is also displayed. If any of the information is incorrect, click the **Back** button to make corrections. If the information is accurate, click the **Next** button to begin installing.

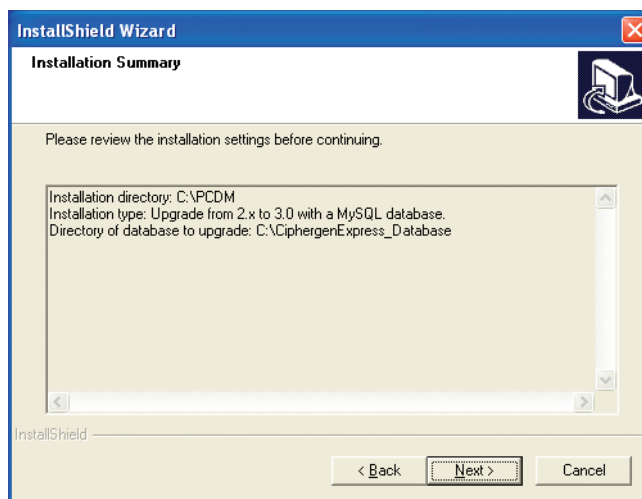


Fig. 5.5: The **Installation Summary** page.

12. Once the files are copied, another installation program will be launched to update the MySQL database tables.

Install the Database

1. After the program files have been transferred, the ProteinChip data manager database installer program will launch automatically. On the first page of the **ProteinChip Data Manager Database Installer** (Fig. 5.6), under **Enter MySQL connection information**, enter the following information:
 - a. Enter the hostname or IP address of the computer that contains the MySQL database ("localhost" should be used if the database is on the same machine to which you are installing the server) into the **Host/IP** field.
 - b. The **Database** should be *CiphergenDM*.
 - c. The **User** should be *root*.

- d. Enter the password for the database. The password is blank by default. However the password will be *cipheradmin* if you are upgrading from a backup copy of your MySQL data folder.

After all the information is entered, click the **Next** button to continue.

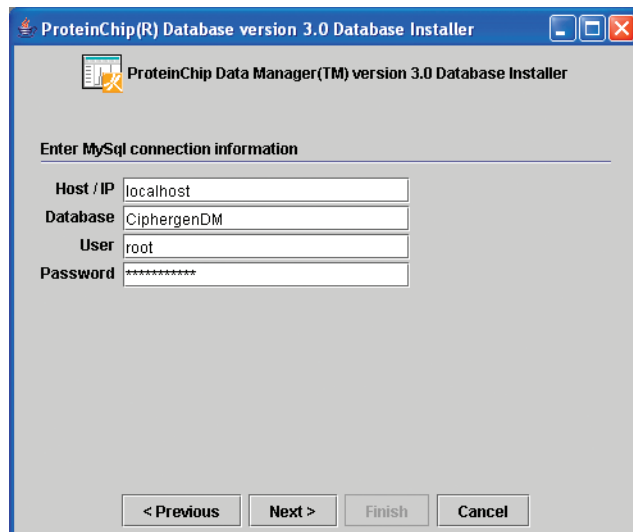


Fig. 5.6: The first page of the **ProteinChip Data Manager Database Installer**.

2. The **ProteinChip Data Manager Install Action** window will display the status of the database table update. After the process is complete, click the **Finish** button.
3. Once the **ProteinChip Data Manager Database Installer** is finished, you will be returned to the main installer program. Click the **Finish** button to complete the installation.
4. Once the server installation is complete, you are ready to install the ProteinChip data manager client software onto the workstation. Proceed to Chapter 7, Installing the Client Software, for further instructions.



NOTE

If any problems or errors were encountered during the installation, please send an email, including any error messages and attached screenshots, to LSG_TECHSERV_US@bio-rad.com.

Chapter 6: Upgrading From Version 2.x Using Oracle

Introduction	47
Before You Begin	47
Back Up the Oracle Database	47
Uninstall the Previous Version of CiphergenExpress Software	48
Upgrade the Server and Database	48
Install the Server Software	48
Update the Database Tables	51

Introduction

This section provides instructions for upgrading CiphergenExpress software version 2.x to ProteinChip Data Manager software version 3.0.7 using an Oracle database. The instructions are the same for upgrading from version 2.0 or 2.1. You cannot upgrade directly from ProteinChip data manager software version 1.0 to version 3.0.7. The upgrade process installs the new server software and updates the database tables to be compatible with new features included in the version 3.0.7 software. If you are not updating from version 2.x using the Oracle database, please see Installation Choices on page 8.

Please email any installation questions to LSG_TECHSERV_US@bio-rad.com.



NOTE

The upgrade can take several hours if the database is very large.



IMPORTANT

The upgrade process requires free disk space equal to a quarter the size of the database. See your Oracle database administrator to ascertain the size of the existing CiphergenExpress database.



WARNING

Follow the instructions below exactly. Failure to do so may result in a failure to migrate the existing data.

Before You Begin

Back Up the Oracle Database

Bio-Rad strongly recommends that you back up the CiphergenExpress database before performing the upgrade procedure. Consult Oracle technical documentation for backup procedures if your ProteinChip data manager database is implemented with Oracle.



NOTE

*You may need to stop the **CiphergenExpress Server** service before performing a backup. To stop the service, select **Control Panel | Administrative Tools | Services**. Right-click **CiphergenExpress Server** and select **Stop** from the pop-up menu.*

Uninstall the Previous Version of CiphergenExpress Software

Before ProteinChip data manager software 3.0.7 can be installed, you must first uninstall the previous version.

1. Uninstall the CiphergenExpress server software. Select **Control Panel | Add/Remove Programs**. Locate the program called **CiphergenExpress Server** and uninstall it. Delete the **CiphergenExpress_Server** folder.
2. Select **Control Panel | Add/Remove Programs**. Uninstall **Java 2 Runtime Environment, SE v1.4.2**. Then, uninstall **Java 2 SDK, SE v1.4.2**.

Upgrade the Server and Database

Install the Server Software

1. Insert the ProteinChip data manager software CD into the CD-ROM drive of the computer to be used as the ProteinChip data manager server. If the program does not start automatically after a few moments, browse to the CD using Windows Explorer and double-click *setup.exe*.
2. At the **Welcome** page, click the **Next** button.
3. At the **License Agreement** page, please read the license agreement. You must click the **Yes** button to accept the agreement in order to proceed.
4. Please read the preinstall instructions on the **Install checklist** page, then click the **Next** button.
5. At the **Choose Destination Location** page, specify where on the hard drive you would like to install the ProteinChip data manager server software. The default path is C:\PCDM. If you must change the path (to change the drive letter, for example), make sure there are no spaces in the path or the software will not function correctly. Click the **Next** button to continue.



IMPORTANT

*If you modify any part of the default path, you **must** ensure that there are **no spaces** in the path name or the software will not run correctly.*

6. At the **Setup Type** page, select **4: Upgrade from v.2x to v3.0.7 with an Oracle database** and then click the **Next** button (Fig. 6.1).

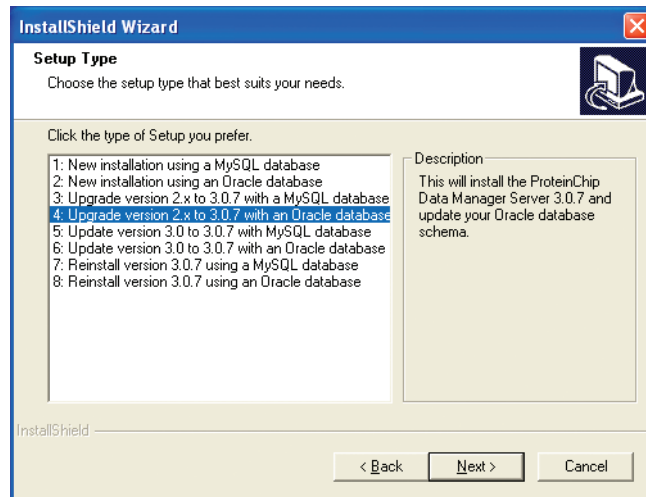


Fig. 6.1: The Setup Type page.

7. At the **Enter Text** page, enter the network name of the computer that is running the Oracle database. If ProteinChip data manager server software and the Oracle database are running on the same machine, this field can be left as "localhost". Click the **Next** button to continue (Fig. 6.2).

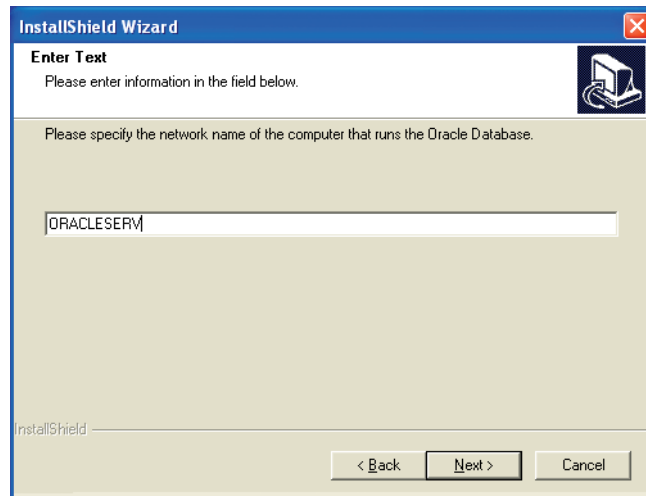


Fig. 6.2: The Enter Text page.

8. At the **Select Program Folder** page, choose the location where the program shortcuts will be placed in the computer's **Start** menu. The default location is **ProteinChip Data Manager Server version 3.0.7**. Click **Next** to continue.
9. The **Installation Summary** page displays summary information about the setup type that was chosen (Fig. 6.3). Please review the contents before proceeding. For new installations using Oracle, the summary should show your specified installation directory, and **Upgrade from 2.x to 3.0.7 with an Oracle database** for the installation type. The name of the Oracle computer should also be displayed. If any of the information is incorrect, click the **Back** button to make corrections. If the information is accurate, click the **Next** button to begin installing. Once the files are copied, another installation program will be launched to create the database tables.

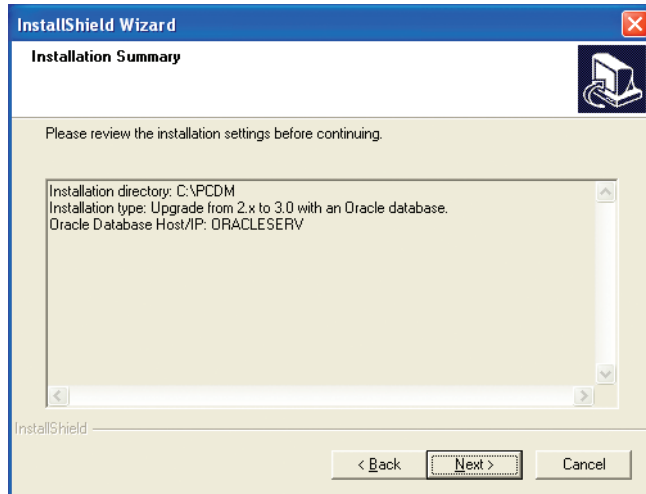


Fig. 6.3: The Installation Summary page.

Update the Database Tables

A separate installation program is used to create tables in the Oracle database.

1. After the program files have been transferred, the ProteinChip Data Manager database installer program will launch automatically. On the first page of the **ProteinChip Data Manager Database Installer**, under **Enter Oracle connection information**, enter the following information:
 - a. Enter the hostname or IP address of the computer that contains the Oracle database ("localhost" can be used if the database is on the same machine upon which you are installing the server) into the **Host/IP** field.
 - b. The **SID/Database** should be *cipherdb*.
 - c. The **User** should be *ciphergen*.
 - d. Enter the **Password** for the database. The default password is *express*.

Once all the information is entered, click the **Next** button to continue (Fig. 6.4).

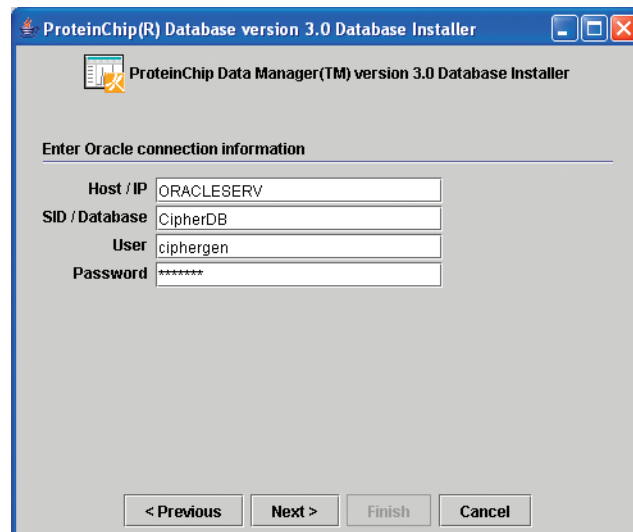


Fig. 6.4: The first page of the **ProteinChip Data Manager Database Installer**.

2. The **ProteinChip Data Manager Install Action** window will display the status of the database table update. Once the process is complete, click the **Finish** button.

3. Once the installation is complete, you are ready to install the ProteinChip data manager client software onto the workstation. Proceed to Chapter 7, Installing the Client Software, for further instructions.

**NOTE**

If any problems or errors were encountered during the installation, please send an email, including any error messages and attached screenshots, to LSG_TECHSERV_US@bio-rad.com.

Chapter 7: Installing the Client Software

Before You Begin	55
Installing and Configuring the Client Software	55
Installing the Client Software	55
Configure the Client Software's Server Connection	56
Register the Server	58

Before You Begin

You must have local administrative privileges on the workstation in order to install the client software. Contact a network administrator for assistance if required.

Please email any installation questions to LSG_TECHSERV_US@bio-rad.com.

Installing and Configuring the Client Software

Installing the Client Software

Once the database and server installation is complete, the client software can be downloaded and installed using a browser. You must have local administrative privileges on the workstation in order to install the client software. Contact a network administrator for assistance if required.

To install ProteinChip data manager client software:

1. Close all other programs.
2. Launch an Internet browser and open the URL
`http://<hostname>:6611/CiphergenExpressDM/install.htm`
where `<hostname>` is the name of the ProteinChip data manager server computer.



NOTE

Enter the URL exactly as shown above — it is case sensitive.

3. The **Warning - Security** dialog box may appear (Fig. 7.1). Click **Yes** to continue the installation. Click **Always** to proceed and hide the dialog box in future.

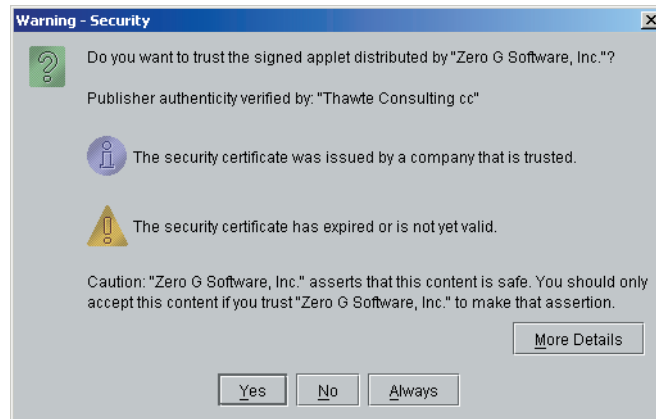


Fig. 7.1: The Warning - Security dialog box.

4. The ProteinChip data manager client software installation web page appears (Fig. 7.2). Click **Start Installer for Windows...** and follow the installation instructions.

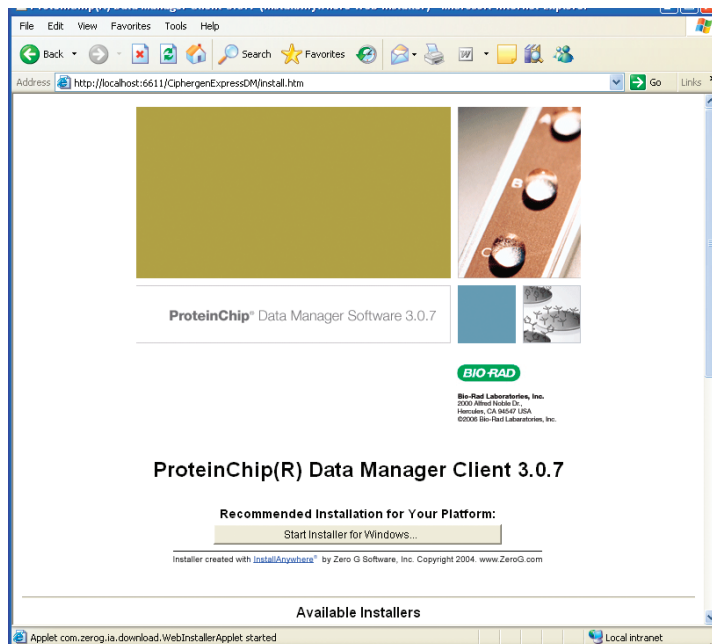


Fig. 7.2: The ProteinChip data manager client software installation web page.

Configure the Client Software's Server Connection

The first time you launch the client software, you must configure its connection to the ProteinChip data manager server software.

1. Start the client software by selecting **Start | Programs | ProteinChip Data Manager | ProteinChip Data Manager Client** or by double-clicking the ProteinChip data manager client icon on the desktop.
2. When the log-in dialog box appears, click the ellipse button (...) next to the **Server** list (Fig. 7.3).

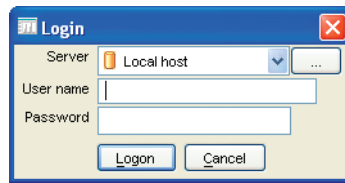


Fig. 7.3: The **ProteinChip Data Manager Log-in** dialog box.

3. When the **Servers** dialog box appears, click **New** (Fig. 7.4).

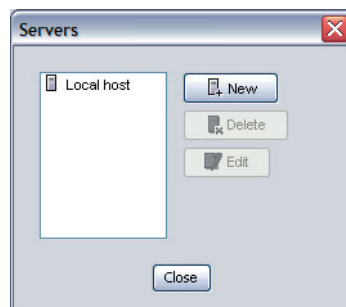


Fig. 7.4: The **Servers** dialog box.

4. The **New server** dialog box appears (Fig. 7.5).

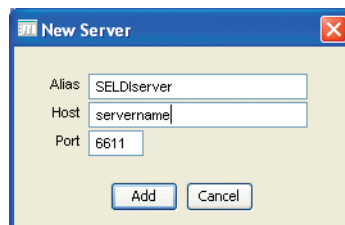


Fig. 7.5: The **New server** dialog box.

5. Enter a nickname for the server in the **Alias** field. Enter the name of the server computer in the **Host** field and make sure the **Port** value is 6611. Click **Add**.
6. The **Servers** dialog box will reappear, listing both your new server and the default "localhost" server. You may wish to delete the "localhost" server if it is not in use. Click **Close**.

Register the Server

1. After you have installed the client software and configured its connection to the server, log in as the administrator. The user name is *admin* and the password is *admin*.
2. If this is the first time you have logged in, the **Register now** dialog box appears, asking if you want to register. If the software was upgraded from 2.x to 3.0.7, the license file should be transferred from the 2.1 version to 3.0.7 automatically during the upgrade process (Fig. 7.6).

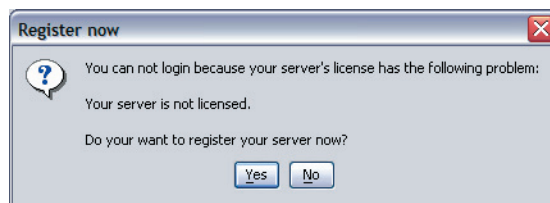


Fig. 7.6: The **Register now** dialog box.

3. Click **Yes**, then enter the location of the license file obtained from Bio-Rad in the dialog box that appears (or use the **Browse** button to browse to it). Once this step is completed you will be able to start the client software (Fig. 7.7).

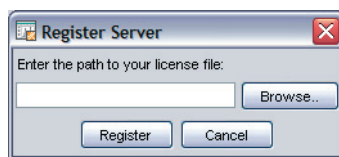


Fig. 7.7: The **Register Server** dialog box.

4. We recommend that you change the *admin* password at this time.

Chapter 8: Troubleshooting

Troubleshooting Installations and Upgrades

61

Contacts

61

Troubleshooting Installations and Upgrades

If the command window flashes by, but the ProteinChip data manager database installer application never opens, continue with the rest of the installation or upgrade, but do not attempt to use the client software. After the installation or upgrade is complete, manually run the ProteinChip data manager database installer application.

1. Stop the server service. To shut down the server, go to **Control Panel | Administrative Tools | Services**. Right-click on **ProteinChip Data Manager Server** and select **Stop** in the pop-up menu.
2. In Windows Explorer, navigate to the **DBUpgrade** folder on the ProteinChip data manager software CD.
3. Run the **dbinstall.bat** file by double-clicking on it.
4. A command window should appear, followed by the ProteinChip data manager database installer application. Note that there may be a time lag before the application opens.
5. Select the operation you are attempting. Refer to the previous instructions for the type of database and installation or upgrade you are performing.
6. If errors occur, contact Bio-Rad immediately at LSG_techsupport_us@bio-rad.com.
7. Restart the server service. To start the server, go to **Control Panel | Administrative Tools | Services**. Right-click on **ProteinChip Data Manager Server** and select **Start** in the pop-up menu.

Once ProteinChip data manager software is installed, if you receive error messages such as "Unable to connect to server at port 6611" check the following:

1. Go to **Control Panel | Administrative Tools | Services** and check that **ProteinChip Data Manager Server** and **MySQL** show **Started** as their status. If they are not started, you can start them by right-clicking and selecting **Start**.
2. If starting the services from the **Control Panel** fails, then reboot the server.
3. If **ProteinChip Data Manager Server** and **MySQL** do not appear in the **Services** panel, check the paths to the locations where the ProteinChip data manager database and the ProteinChip data manager server software are installed. If there are any spaces in the path names, you must uninstall both the database and server software and reinstall ensuring that there are no spaces in any path names.

Contacts

For all technical, licensing, and software questions, please contact LSG_TECHSERV_US@bio-rad.com.

Index

C

- client software
 - configuring server connection 56–57
 - installation 55–56
 - URL for downloading 55
- client-server configurations
 - PBS-II instruments 7
 - ProteinChip SELDI systems 6
- compatibility requirements for ProteinChip software 6

D

- database installation choices 9

I

- installation choices 9
- installing
 - new MySQL database and server 13–15
 - new Oracle database and server 19–25
 - troubleshooting installation 61

L

- license (ProteinChip data manager server software), obtaining 5, 8
- logging in 61

M

- MySQL
 - installing new database and server 13–15
 - upgrading version 2.x database and server 39–44
 - upgrading version 3.0 database and server 29–31

N

- New server** dialog box 57

O

- Oracle
 - installing new database and server 19–25
 - upgrading version 2.x database and server 47–52
 - upgrading version 3.0 database and server 32–35

P

PBS-II instruments, client-server configuration 7

ProteinChip data manager software

- components 3

- obtaining server license 5, 8, 13, 19

- registering server 58

ProteinChip SELDI system, client-server configuration 6

ProteinChip software, compatibility requirements for ProteinChip data manager software 6

- using version 3.0 and MySQL database 29–31

- using version 3.0 and Oracle database 32–35

- URL for downloading client software 55

R

Register now dialog box 58

Register Server dialog box 58

registering ProteinChip data manager server software 58

S

software installation, see *installing*

software upgrades, see *upgrading*

system requirements

- client workstation 5

- server 4

U

upgrading

- using version 2.x and MySQL database 39–44

- using version 2.x and Oracle database 47–52



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